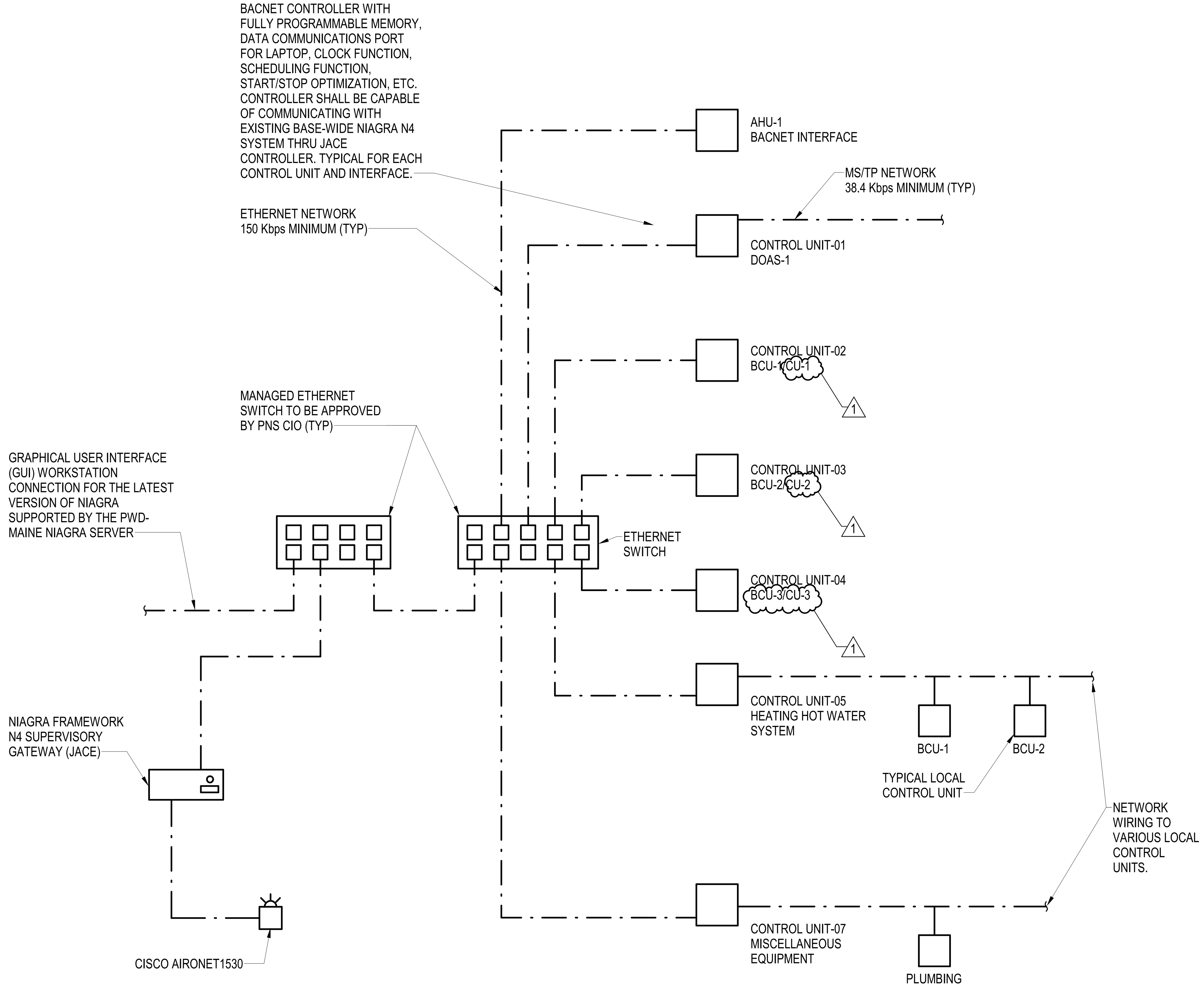
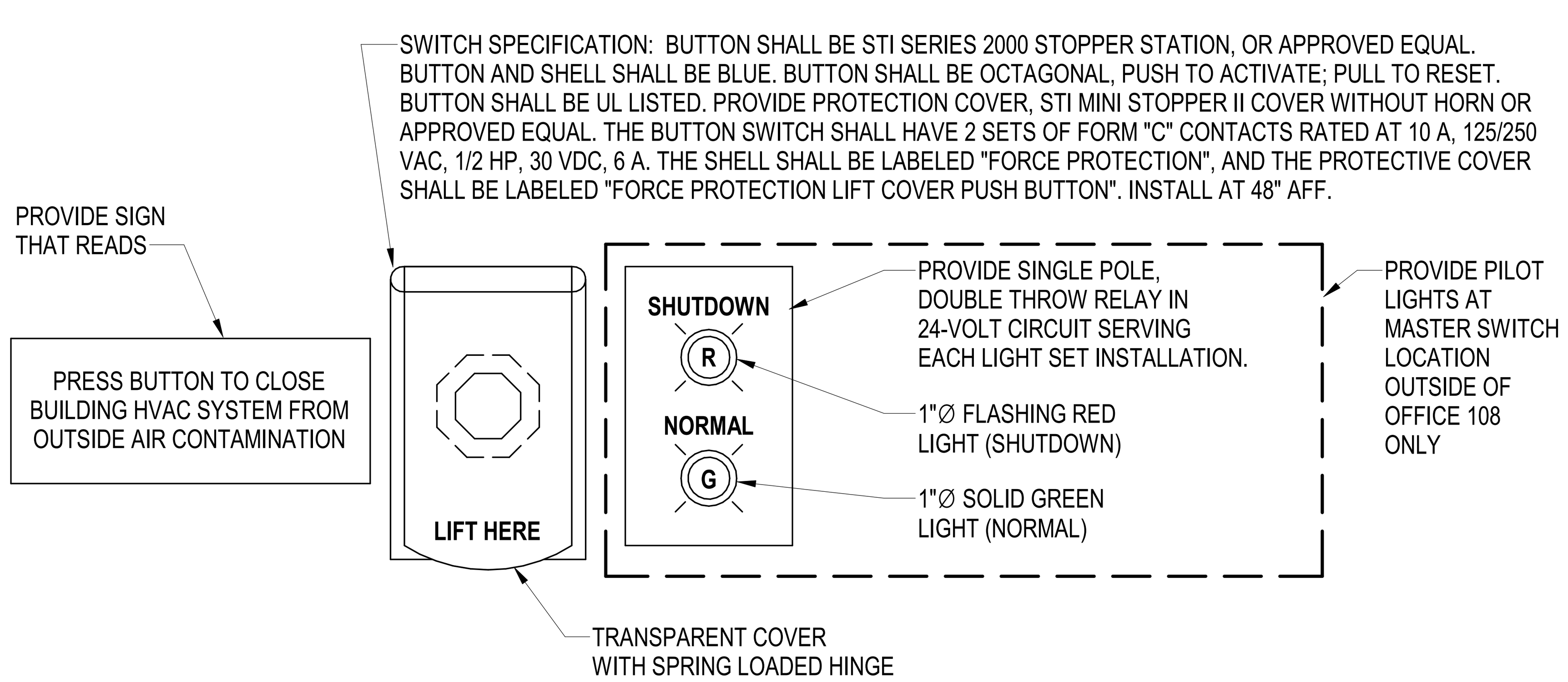


B1 DIRECT DIGITAL CONTROL SYSTEM ARCHITECTURE

SCALE: NOT TO SCALE

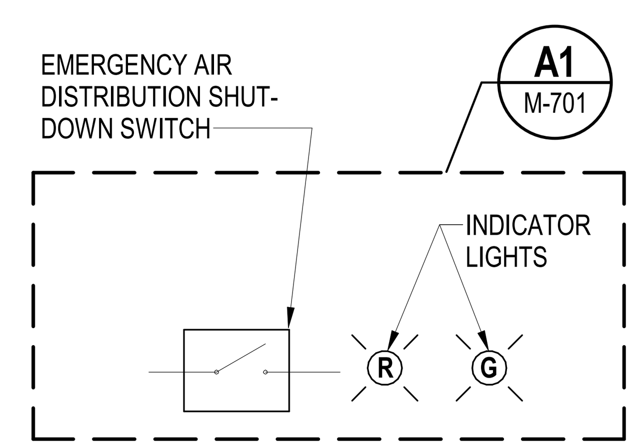


A1 EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH (EAS) DETAIL

SCALE: NOT TO SCALE

CONTROL SYSTEM GENERAL NOTES

1. PROVIDE A COMPLETE OPERATIONAL DIRECT DIGITAL CONTROL (DDC) SYSTEM IN ACCORDANCE WITH THE PROJECT SPECIFICATIONS AND DRAWINGS. THE SYSTEM SHALL CONSIST OF NATIVE BACNET DEVICES AT ALL LEVELS COMMUNICATING VIA ASHRAE 135 BACNET PROTOCOL OVER APPROPRIATE BACNET LISTED LANS. WHEREVER POSSIBLE, LIMIT THE USE OF SENSORS AND DEVICES WHICH COMMUNICATE IN IP PROTOCOL OVER IP LANS. WHEN THIS IS NOT POSSIBLE, PLACE ALL IP LAN CABLE IN METAL CONDUIT WITH METALLIC LOCKABLE ENCLOSURES AT BOTH ENDS. THE DDC SYSTEM SHALL CONNECT TO A NIAGARA SUPERVISORY GATEWAY (COMMONLY REFERRED TO AS A JACE). PROVIDE A BUILDING WORKSTATION LAPTOP COMPUTER WITH NIAGRA FRAMEWORK SOFTWARE VERSION 4.2 OR LATER THE BUILDING WORKSTATION SHALL INCLUDE NIAGARA WORKBENCH OR EQUIVALENT NIAGARA FRAMEWORK ENGINEERING TOOL SOFTWARE. THE CONTRACTOR SHALL CREATE SYSTEM GRAPHICS, DATA TRENDING, DATA LOGGING, ALARMING AND TROUBLESHOOTING FOR THE DDC SYSTEM. THE CONTRACTOR SHALL DISCOVER THE DDC SYSTEM GRAPHICS VIA THE JOIN PROCESS IN THE BUILDING 374 BASE-WIDE EMCS. THE FACILITY POINT OF CONNECTION SHALL BE A REPLACEMENT CISCO AIRONET 1530 WIRELESS ACCESS POINT. PROVIDE LICENSE FOR NIAGRA JACE TO SUPPORT THE NUMBER OF NETWORK DEVICES REQUIRED FOR THIS PROJECT. LICENCE FOR JACE AND LAPTOP IS TO INCLUDE 5 YEARS OF SOFTWARE, UPDATES AND WARRANTY.
2. ALARMS SHALL BE ANNUNCIATED ON THE GRAPHICAL USER INTERFACE (GUI).
3. SETTINGS, MODES AND SET POINTS, THAT ARE INDICATED AS BEING ADJUSTABLE, SHALL BE ADJUSTABLE BY THE SYSTEM OPERATORS THROUGH POINT-AND-CLICK ICONS ON THE GUI WITHOUT THE NEED TO CHANGE OR EDIT PROGRAMMING.
4. ANALOG DATA SHALL BE TRENDED AT REGULAR INTERVALS, DETERMINED BY THE EXPECTED RATE OF CHANGE OF THE DATA, AND SHALL BE ARCHIVED AND STORED ON THE GUI COMPUTER. TREND DATA SHALL BE CONFIGURED AND STORED TO SATISFY NAVY REQUIREMENTS FOR FUTURE MEASUREMENT AND VERIFICATION.
5. BINARY DATA SHALL BE TRENDED ON A CHANGE OF STATE BASIS AND SHALL BE ARCHIVED AND STORED ON THE GUI COMPUTER. TREND DATA SHALL BE CONFIGURED AND STORED TO SATISFY NAVY REQUIREMENTS FOR FUTURE MEASUREMENT AND VERIFICATION.
6. PROVIDE LOCAL CONTROL UNITS AS REQUIRED TO CONTROL BLOWER COIL UNITS OR OTHER TERMINAL EQUIPMENT .
7. CONTROL CONTRACTOR SHALL PROVIDE DDC/UMCS SYSTEM WITH TWO UNIQUE SETS OF ACCESS PASSWORDS. ONE SET OF PASSWORDS, PRIMARY USER, SHALL BE FOR NAVFAC PWD-ME SHOP PERSONNEL WITH COMPLETE ACCESS TO ALL POINTS, GRAPHICS AND PROGRAMMING. THE SECOND SET OF PASSWORDS, SECONDARY USER, SHALL BE FOR LIMITED ACCESS BY CODE 980 FOR SUPPORT AND MAINTENANCE OF SPECIFIC SYSTEMS AND/OR EQUIPMENT. LIST OF SECONDARY USER POINTS AND ACCESS SHALL BE SUBMITTED TO CONTRACTING OFFICER FOR APPROVAL.
8. COMPLY WITH CYBERSECURITY REQUIREMENTS AND PERFORM REQUIRED PROGRAMMING AND TESTING PER SPECIFICATIONS.
9. ALL NIAGRA INTEGRATION SHALL FOLLOW THE LATEST VERSION OF THE NAVFAC PWD-ME ICS STANDARDS.
10. BAS CONTRACTOR SHALL PROVIDE A COMPLETE CONTROL POINT MATRIX INDICATING THE FUNCTION OF THE INDIVIDUAL COMPONENTS OF THE BUILDING AUTOMATON. THE BAS CONTRACTOR SHALL SUBMIT A DETAILED NARRATIVE INDICATING THE SPECIFIC FUNCTION OF EACH COMPONENT AND DEVICE. THE NARRATIVE SHALL NOT BE A REPLICATION OR DUPLICATION OF THE ENGINEER'S SEQUENCE OF OPERATION.
11. THE BAS SUBCONTRACTOR SHALL PROVIDE UTILITY MONITORING OR ELECTRICAL USAGE AND DEMAND FOR THE CHILLED WATER SYSTEM (IE:CHILLERS, PUMPS, ETC.).
12. THE BAS SUBCONTRACTOR SHALL ASSIST THE BALANCING CONTRACTOR WITH THE BALANCING AS WELL THE COMMISSIONING AGENT IN SYSTEM VERIFICATION.
13. PRIOR TO FINAL COMPLETION, THE BAS CONTRACTOR SHALL DEMONSTRATE TO THE COORDINATE OFFICE AND ENGINEER THAT ALL SYSTEMS ARE FUNCTIONING AS DESIGNED AND MEETS THE INTENT OF THE DRAWINGS AND SPECIFICATIONS.



SEQUENCE OF OPERATION

IN THE EVENT THAT THE EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH IS ACTIVATED, THE FANS IN AIR HANDLING EQUIPMENT SHALL STOP AND THE OUTSIDE AND EXHAUST AIR DAMPERS ASSOCIATED WITH THESE UNITS SHALL CLOSE. THE FANS SHALL REMAIN STOPPED AND THE DAMPERS SHALL REMAIN CLOSED UNTIL THE SWITCH IS RELEASED. REFER TO EQUIPMENT SPECIFIC CONTROL DIAGRAMS FOR FORCE PROTECTION SHUTDOWN PROTOCOLS.

EADS POINTS LIST

SYSTEM POINT DESCRIPTION	GRAPHIC	ANALOG INPUT	ANALOG OUTPUT	BINARY INPUT	BINARY OUTPUT	ALARM	ANALOG VARIABLE	BINARY VARIABLE	TREND LOG		NOTES
EM. AIR DISTRIBUTION SHUTDOWN SWITCH	X			X		X			X		1,2,4
SHUTDOWN LIGHT (RED)	X				X				X		3
NORMAL LIGHT (GREEN)	X				X				X		3

NOTES:
1. GENERATE AN ALARM IF THE EMERGENCY AIR DISTRIBUTION SHUTDOWN SWITCH IS CLOSED.
2. TYPICAL OF 3. REFER TO SHEETS MP102, MP104, AND MP105 FOR LOCATIONS.
3. PROVIDE AT SINGLE LOCATION ONLY. REFER TO SHEET MP102.
4. PROVIDE "VIRTUAL" EADS SWITCH ON GUI FOR REMOTE OPERATION AND TESTING THROUGH DDC SYSTEM.

A3 EMERGENCY AIR DISTRIBUTION CONTROL DIAGRAM

SCALE: NOT TO SCALE

MARK	DESCRIPTION	DATE	APPR
1	AMENDMENT 008	03/10/2026	RSH
	FINAL SUBMISSION	10/01/2025	RSH

NAVFAC

STATE OF MAINE RICHARD CRESWELL No. 13178 LICENSED PROFESSIONAL ENGINEER 10-01-25

SEAL

OAK POINT ASSOCIATES

APPROVED

FOR COMMANDER NAVFAC

ACTIVITY

SATISFACTORY TO DATE

DESIGNER RNC

DRAWN BY CBM

CHECKED BY MSA

DESIGN MANAGER T. CARLETON

PROJECT MANAGER L. LYONS

FEAD/PM&E

FIRE PROTECTION S. RUCINSKI

DEPARTMENT OF THE NAVY NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND
NAVAL FACILITIES ENGINEERING SYSTEMS COMMAND - MID-ATLANTIC
ROICC-ME PORTSMOUTH NAVAL SHIPYARD - KITTERY, ME
PORTSMOUTH NAVAL SHIPYARD KITTERY, ME

B60 REPAIRS

MECHANICAL CONTROL DIAGRAMS 1

EPROJECT NO.: 1740579

NAVFAC DRAWING NO.

SHEET 220 OF 282

M-701

DRAWFORM REVISION: 14 SEP 2023

UNCLASSIFIED

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